

EEG Feature Extraction of Motor Imagery Based on WT and STFT*

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Abstract - Aiming at the problem of low recognition rate in classification task of motion imagery, an EEG feature extraction algorithm based on Wavelet Transform (WT) and Short-Time Fourier Transform (STFT) is proposed. Firstly, the wavelet decomposition of the EEG on a specific lead is used to reconstruct the frequency containing the Event-Related Desynchronization / Synchronization (ERD / ERS) features to remove the redundant information. Then the Short-Time Fourier Transform is used to extract the feature of the motor imagery, and visualized these features. Finally, the convolution neural network is used to classify and obtain the final result. The proposed method is applied to EEG data of two kinds of motion imaging tasks in BCI competition, and the experimental results show that the recognition rate of classification can reach 96.67% and the average Kappa coefficient is 0.93, which verifies that our proposed algorithm can effectively distinguish two types of motor imagery task, and improve the recognition rate of classification.

Index Terms - Motion Imagery; Wavelet Transform; Short-Time Fourier Transform; Convolution Neural Network.

I. INTRODUCTION

Brain-Computer Interface (BCI) is a technology that directly connects the human brain to external devices without relying on traditional neural pathways. As a new interdisciplinary subject, BCI technology has attracted the attention of many domestic and international scientists and has become a hot research topic in the fields of biomedicine, communication and computer technology [1]. Brain-computer interface based on Motor Imagery (MI) is a very important brain-computer interaction strategy, and the effective feature extraction method is great importance to improve classification accuracy [2].

Classical EEG signals of motor imagery generally used single type of feature extraction methods such as autoregressive (AR) model [3], correlation coefficient analysis [4], wavelet transform [5], power spectrum estimation [6], etc. The accuracy of the AR model is affected by the length of the data segment. The shorter the data length, and the higher the temporal resolution, and the greater the error of the model estimation. By using the correlation coefficient analysis, the correlation coefficient can be directly selected from the original features and the task correlation coefficient is larger of the parameters to avoid the loss of the original information, but the calculation is large, and the calculation results are not precise enough. In reference [7], Multiple Autocorrelation Analysis

(MAA) are proposed to extract EEG signals. In reference [8], convolutional neural networks are directly used as a classifier to classify signals. In reference [9], a method which combine Wavelet Transform and Sample Entropy (SE) to extract features. WT is a typical time-frequency analysis method in the time and frequency domain which have good resolution, and suitable for analysis non-stationary signal. The power spectrum estimation belongs to the frequency domain analysis method, which can't well characterize the time-domain features of the signal. Because different analysis methods have their own characteristics and limitations, it is often difficult to extract the feature information of all non-stationary signals with only one analysis method, and the information obtained is not enough.

In recent years, many scholars tried to combine two feature extraction methods, and got good recognition results on the data set III of BCI competition. In reference [10], a method combine wavelet packet and ICA was used to obtain the motion rhythm associated with the ERD / ERS phenomenon, with a maximum recognition rate of 88.9%. In reference [11], the feature extraction method based on the combine ERD band energy features and cumulative energy features was used to construct the joint eigenvector, which achieved the maximum classification recognition rate of 90%. In reference [12], two features were extracted using autoregressive model and wavelet transform method, and used genetic algorithm for optimal feature set selection, and the recognition rate can reach of 96.43%.

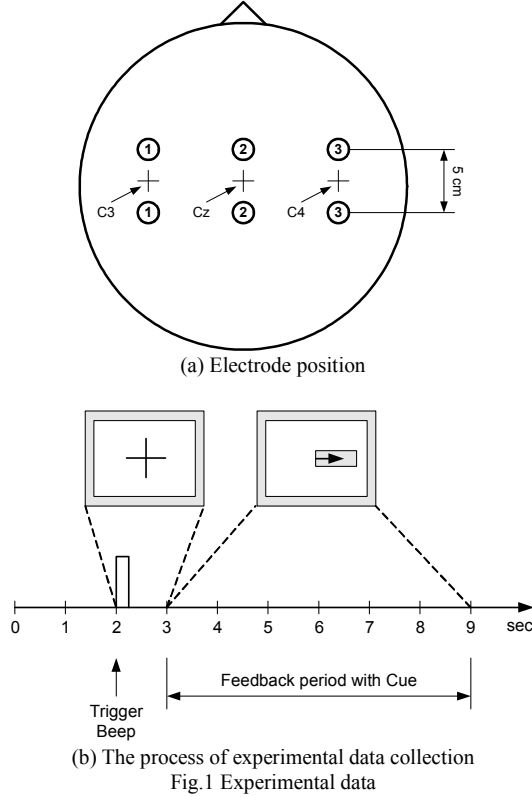
Because WT can highlight the characteristics of signal specific frequency band, and STFT can clearly show the signal frequency changes with time, so this paper presented an extraction method which combine WT with STFT, and the convolution neural network is used to classify. The method effectively improved the accuracy of motor imagery in EEG signals of MI, and provided a new idea for EEG feature extraction methods.

II. DATASET SELECTION

The experimental data were derived from the data set III of the 2003 BCI competition dataset published by the Institute of Bioengineering, Graz University of Science and Technology, Austria. Experimental data from a 25-year-old healthy women, which is collect experiments divided into 7 groups, and each group of 40 experiments, and imagine the right hand movement of 140 times, and a total of 280 experiments. Among them, 140

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experimental data were used as training samples and another 140 samples were used as test samples. Experimental task for the subjects according to the cursor left / right direction, and imagine the corresponding left / right hand movement, EEG data from the differential electrode from the international standard 10-20 lead system C3, Cz and C4 three channel acquisition, sampling frequency 128 Hz, and after 0.5 ~ 30 HZ filter. Each acquisition signal process shown in Figure 1. Acquisition time is 9s and the first 2 seconds of the subject was relaxed, and the second the system voice prompts the experiment began, the computer screen appears the symbol "+" to remind the subject to focus, and the 3 second when the screen will appear an arrow (left or right) . Subjects should imagine left / right hand movement according to the direction of the arrow.



III. FEATURE EXTRACTION AND CLASSIFICATION

People in different limb movements or imagination exercise, and the brain sensory motor area will produce a specific EEG signals, which is in the process of discharge of a large number of neurons, by the postsynaptic potential caused the sum of the extracellular field potential [13]. MI refers to the fact that a person carries out the limb movement imagination without actual limb movement. The perceived rhythm consists of μ and β rhythms, and they are the fluctuations of the brain activity at 7-13 Hz and 19-26 Hz [14]. When one person is thinking unilateral limb movements, for example imagining left-handed movements, there is "Event-Related Desynchronization (ERD)" in the contralateral and right cerebral cortex regions, with μ or β rhythms decreasing or disappearing, and there is an "Event-Related Synchronization (ERS)" phenomenon in the regions of the ipsilateral and left cerebral cortex that increase μ and β rhythms [15]. Therefore, highlighting the energy changes in μ and β rhythms can be used as a breakthrough to distinguish between two types of imagination.

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A. Discrete Wavelet Transform

EEG signal is a typical non-stationary signal, and only extract the characteristics of the time domain or frequency domain analysis, and cannot get all the information EEG. Wavelet analysis is a multi-resolution time-frequency signal analysis method, with zooming characteristics, which can analyze nonstationary time-varying signals in the frequency and time domains simultaneously [16]. The process of decomposing the original signal by the low-pass and high-pass filters is the process of wavelet decomposition, and the formula is as follows:

$$W_\varphi(j_0, k) = \frac{1}{\sqrt{M}} \sum_n f(n) \varphi_{j_0, k}(n) \quad (1)$$

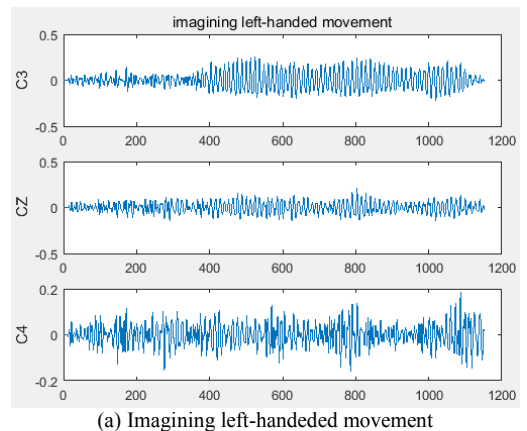
$$W_\psi(j, k) = \frac{1}{\sqrt{M}} \sum_n f(n) \psi_{j, k}(n), j \geq j_0 \quad (2)$$

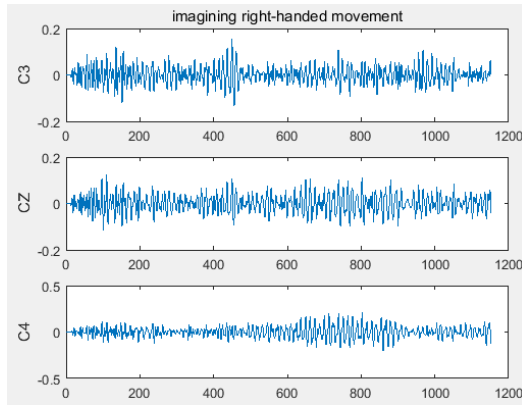
Where j is the scale, j_0 is the initial scale, $\varphi(n)$ is the scale function, and it is used to control the scaling of the wavelet function $\psi(n)$, where k is the translation. For a signal $f(n)$, $W_\varphi(j_0, k)$ is an approximate coefficient obtained by wavelet decomposition, and $W_\psi(j, k)$ is a detail coefficient. The Multi-layer decomposition of WT is continue to decompose the low-frequency components after one layer of decomposition, and high-frequency components will no longer continue to decompose.

Wavelet decomposition of multi-level signal, the band under different scales signal. Reconstructing the wavelet coefficients of the corresponding frequency bands can obtain signals containing only certain frequency segments. The reconstruction formula is as follows:

$$f(n) = \frac{1}{\sqrt{M}} \sum_k W_\varphi(j_0, k) \varphi_{j_0, k}(n) + \frac{1}{\sqrt{M}} \sum_{j=j_0}^{\infty} \sum_k W_\psi(j, k) \psi_{j, k}(n) \quad (3)$$

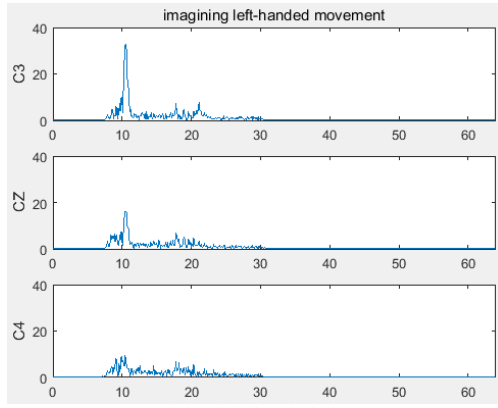
In this paper, 140 sets of training samples and 140 sets of test samples in the data set are subjected to discrete wavelet transform. The db4 wavelet base is selected to decompose the original waveform by three layers of wavelet, and obtained high-frequency coefficients of 16-32HZ and 8-16HZ combining and reconstructing, the waveform after reconstruction shown in Figure 2.



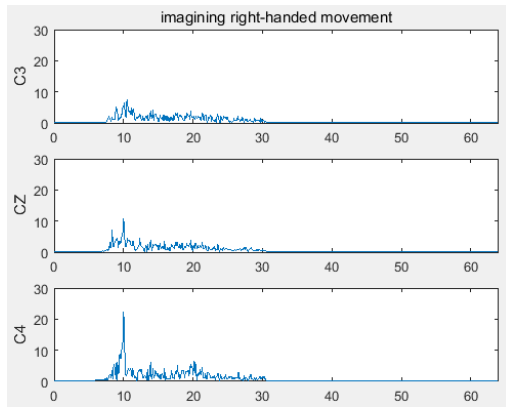


(b) Imagining right-handed movement
Fig.2 The wave chart of MI

The spectrogram shown in Figure 3, we can see that the wavelet decomposition and the waveform of the specified frequency segment reconstruction, and you can get the effect of filtering to remove redundant information.



(a) Imagining left-handed movement



(b) Imagining right-handed movement
Fig.3 Spectrogram of MI

B. Short-Time Fourier Transform

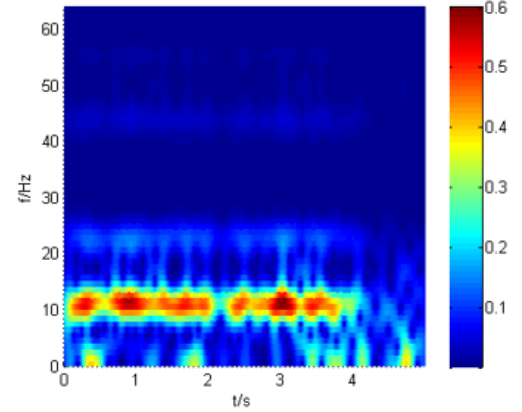
Short-Time Fourier Transform is based on the Fourier Transform to the signal plus a window function. By moving the window function, the signal is divided into different time periods, and then the signal for each time period Fourier transformed, and the signal in each time period can be approximated as a stationary signal. The formula for short-time

Fourier transform is as follows:

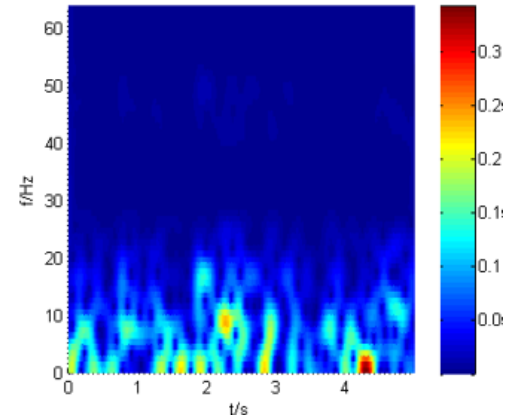
$$X(e^{j\omega}) = \sum_{m=-\infty}^{+\infty} [x(m)w(n-m)]e^{-j\omega m} \quad (4)$$

$x(m)$ is represented original signal and $w(n-m)$ is the window function.

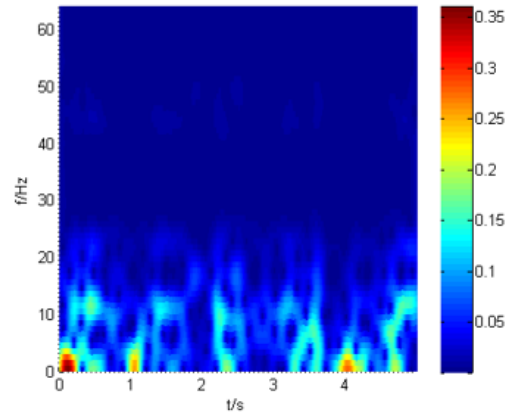
The EEG signal of MI is subjected to STFT with a sampling frequency of 128 Hz. The window function is a Hamming window with a step size of 64, resulting in a time-frequency spectrum with the abscissa being the time and the ordinate being the frequency. As time changes, the frequency characteristics of the imagination movement appear in the energy value.



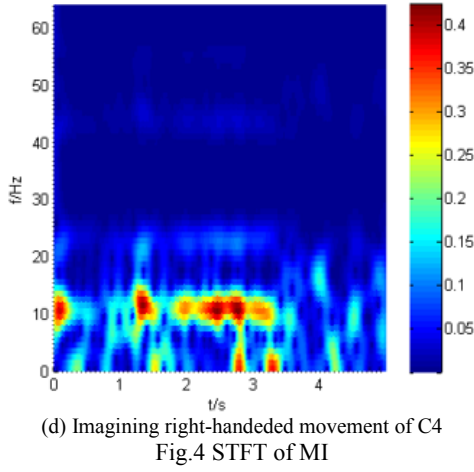
(a) Imagining left-handed movement of C3



(b) Imagining left-handed movement of C4



(c) Imagining right-handed movement of C3



As shown in Fig. 4, comparing the graphs (a) and (b), when the subject imagined left-hand movement, the energy of the whole C3 channel on the left side of the brain was significantly higher than that of the C4 channel on the right side of the brain. Contrast (c) and (d), when the subject imagined right-hand movement, the energy of the C4 channel on the right side of the brain was significantly higher than that of the C3 channel on the left side of the brain. This fulfills ERD / ERS experimentally. After STFT of the signal, the characteristic differences of imagining unilateral hand movements are enlarged, which has laid a good characteristic foundation for classification and recognition.

C. Feature Data Visualization

Visualization refers to the use of computer graphics and image processing technology, and the data into graphics or images displayed on the screen technology. After data visualization, we can use the image processing method for processing. After short-time Fourier transform, the signals of C3 and C4 are obtained respectively. The features of the two channels C3 and C4 are arranged up and down to form an image.

Since the value of its characteristic value is smaller in the interval from 10^{-6} to 10^{-2} , the value is expanded to the

interval [0,255] corresponding to the gray value of the image. The formula as follows:

$$m = \frac{n - \min(n)}{\max(n) - \min(n)} \times 255 \quad (5)$$

The letter n indicated the value of each data feature, and $\min(n)$ is the minimum of the data values, and $\max(n)$ is the maximum of the data values.

D. The classification of Convolutional neural network

Convolutional neural network structure used in this paper, and it is shown in Figure 5. It consists of one input layer, three convolution layers, three down-sampling layers, one fully connected layer and one output layer. The convolution layer and the down-sampled layer each consist of several feature maps. Each point in the feature map represents a neuron. The convolution kernel consists of a trainable weight matrix. The input image is convolved with the convolution kernel operation, and plus bias and through a stimulus function to get the corresponding characteristics of the map, as the formula:

$$y = f(x * w + b) \quad (6)$$

Where x is the input image, w is the weight, b is the offset, and $*$ is the convolution operation.

The down-sampling layer can not only reduce the dimension of the image, and reduce the complexity of the network, but also make an aggregated statistic on the different positions of the convolution layer feature map. The output layer used Softmax regression with strong non-linear classification as a classifier. Given an input picture, the probability that it belongs to a certain category can be expressed as:

$$P = \text{softmax}(\sum_j W_i X_j + b_i) \quad (7)$$

W_i is the weight, b_i is the offset of class, i and j is the pixel index of picture x for pixel summation.

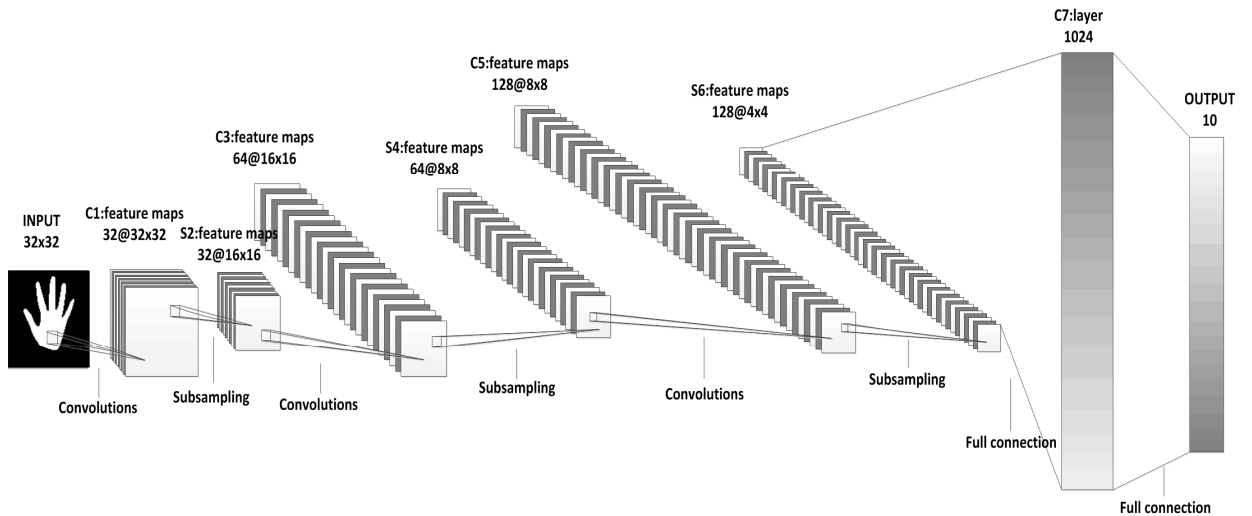


Fig. 5 The structure of convolutional neural networks

The feature-visualized image is used as the input image of the convolutional neural network. The convolution kernel of the network adopts a convolution kernel of 5×5 size, and the down-sampled layer adopts the maximum size of 2×2 samples, and the first several layers of the network select the RELU function as the activation function, and the last output layer adopts the Sigmoid activation function, that is

$$f(x) = \frac{1}{1+e^{-x}} \quad (8)$$

The network structure weight and bias use a smaller positive number to initialize, in order to avoid neuron contact output constant 0 problem. The crossover entropy function is selected as the loss function $J(\theta)$ for specifying the minimum error in the training process, and the loss function is minimized by solving the batch gradient descent method. To reduce over-fitting, the network enables the "dropout" layer during training and shuts down during testing. The batch block size is 30 and the learning rate is constant at 0.001.

IV. EXPERIMENTAL RESULTS

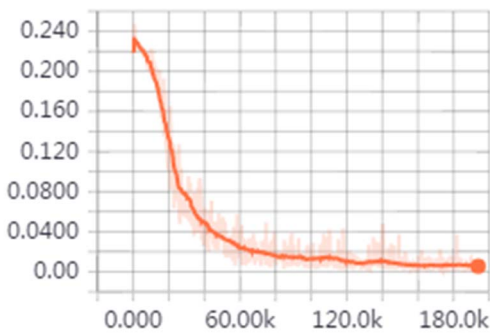
This article used the Tensorflow artificial intelligence learning system developed by the Google Machine Intelligence Institute and Matlab 2016b in Windows as the experimental environment. Tensorflow is an open source, numerical library for software that runs on GPUs based on user-generated flowcharts for fast computation. The computer used was an Intel i5-2400 with 8GB of memory and the graphics card was GTX 1050 Ti with GPU. The experimental feature extraction is completed in Matlab environment, Tensorflow is used to run and test the classifier.

Recognition rate is a commonly used indicator of a good or bad recognition algorithm. Among them, the left hand, right hand movement of the brain imaging classification rate was:

$$\text{The recognition rate}(\%) = \frac{\text{the number of task categories correctly}}{\text{the actual number of imagined exercise}} \times 100\% \quad (9)$$

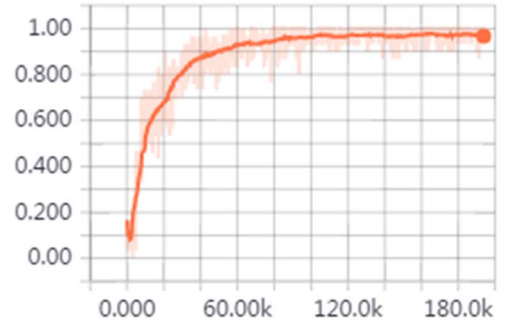
Fig. 6 shows the classification result of convolution neural network by Tensorflow. From the training loss curve we can see, as the number of iterations increases, the loss function decreases and eventually approaches zero. The test accuracy is increased with the increase of the number of iterations, and finally reaching an average accuracy of 96.67%.

xent/loss



(a) Training Loss Curve

test/accuracy



(b) Test Accuracy

Fig. 6 The result of the classification

The kappa coefficient is also a commonly used indicator to evaluate the performance of BCI system [15]. The correctness of random classification is 50% for both types of imagination and 0 for kappa. The *kappa* coefficient is calculated as:

$$kappa = \frac{D - \frac{1}{C}}{1 - \frac{1}{C}} \quad (10)$$

D is the classification accuracy, and C is the number of categories.

Table I shows the recognition results of the two algorithms proposed in this paper and the results of each algorithm separately. It can be seen that the recognition rate achieved by the combination algorithm proposed in this paper is much higher than using either algorithm alone.

TABLE I
THE RECOGNITION RATE

Feature extraction algorithm	The recognition rate (%)	The kappa coefficient
WT	88.57	0.77
STFT	70	0.4
Our method	96.67	0.93

Table II shows the comparison between the feature extraction methods proposed by researchers in the BCI competition dataset in recent years and the methods proposed in this paper. Compared with these feature extraction methods in the same data set, the proposed method has a higher recognition rate and easy applied into practical application of BCI system.

TABLE II
THE RECOGNITION RATE OF COMPARISON

Feature extraction methods	The recognition rate (%)	The kappa coefficient
MAA ^[7]	87.5	0.75
CNN ^[8]	90.75	0.82
WT+SE ^[9]	90.3	0.81
ERD + energy features ^[12]	96.43	0.92
Our method	96.67	0.93

V. CONCLUSION

How to extract the most relevant features of classification tasks and improve the recognition rate has become a hot research topic in the motor imagery class task of EEG. Based on the advantages and problems of the existing feature extraction methods, this paper analyzed the signal from the time domain and the frequency domain by using the characteristics of Wavelet Transform zooming to extract the corresponding frequency and using the Short-Time Fourier Transform to highlight the ERD / ERS phenomenon. Using convolutional neural network with associative memory function for classification. The validation of the BCI data set in 2003 and the comparison with other methods applied in this dataset validate the effectiveness of the proposed algorithm. The experimental results show that the ERD / ERS obtained by the proposed method is more obvious, and the task of motor imagery classification can be better distinguished, and the classifier has higher intelligence. Therefore, the proposed method improves the recognition rate and effectively solves the problem of low recognition rate in the task of motion imaging classification, which provides a new idea for the analysis of EEG signals.

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